

SCm: The Hidden Element — Undetectability, Zero Quantum Signature, and Quasar Ignition

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PAPER_410 – SCm: The Hidden Element — Undetectability, Zero Quantum Signature, and Quasar Ignition Mechanism ****Author:** Daniel T. Murphy** ****Date:** 2025**

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Session: 110 (grok_{share_755feea7}.txt analysis)
CP4 Class: SCmHiddenElementUndetectableQsQuasarIgnitionCalculator (#60)

Abstract

This paper presents a UQFF analysis of SCm: The Hidden Element — Undetectability, Zero Quantum Signature, and Quasar Ignition Mechanism, deriving compressed field equations and observational predictions within the Star-Magic/UQFF framework.

1. Overview

PAPER_410 formalizes the properties of **SCm (Superconductive Material)** as the hidden element central to the UQFF framework. SCm is:

1. **Bound within every atom and star** — donated from stars to planets during system creation
2. **Undetectable by conventional means** — its density eliminates any detectable quantum signature
3. **Exclusively interactive with Ug3** — within planetary cores, exclusively coupled to the magnetic strings disk
4. **The most reactive substance in the universe** — under trapped conditions, also the fastest-moving substance
5. **The fuel of quasar jets** — when Ug1+Ug2+Ug3 fail to trap SCm, it is expelled and ignites against unbound Universal Aether

This paper establishes the **zero-Qs property** of SCm and the **quasar ignition mechanism** as distinct from the heliosphere or planetary core physics.

2. Core Properties of SCm

2.1 Zero Quantum Signature Property

SCm has no detectable quantum signature:

$$Q_s(\text{SCm}) = 0$$

where Q_s is the quantum signature observable. The mechanism:

$$\rho_{\text{SCm}} \sim 10^{15} \text{ kg/m}^3 \quad \text{stellar core}$$

$$\rho_{\text{SCm,planet}} \sim 10^{11} - 10^{13} \text{ kg/m}^3 \quad \text{planetary core}$$

The extreme density causes all quantum field interactions to be self-screened — effectively making SCm **dark** to quantum detectors while remaining physically active.

2.2 SCm Stellar Donation to Planets

During stellar system formation, the parent star donates SCm to each forming planetary body:

$$\text{SCm}_{\text{planet}} = f_{\text{donate}} \cdot \text{SCm}_{\text{star}} \cdot \frac{V_{\text{planet}}}{V_{\text{star}}}$$

where f_{donate} is the fractional donation efficiency. The donated SCm + trapped Universal Aether (UA) in planetary cores is **exclusively interactive with Ug3**:

$$\text{Core interaction rule: } [\text{SCm}]_{\text{planet}} + [\text{UA}] \rightarrow \text{Ug3 only} \quad \text{planetary orbit stabilization}$$

No external interaction:

$$P_{\text{SCm, external}} = 0, \quad P_{\text{SCm, Ug3}} = 10^{-3}$$

2.3 SCm Superconductivity

SCm's superconductivity enables near-lossless magnetic strings:

$$\gamma_{\text{SCm}} = 0.00005 \text{ day}^{-1} \quad \text{(near-zero reciprocation decay)}$$

This gives SCm-driven magnetism (U_m) lifetime:

$$\tau_{\text{SCm}} = \frac{1}{\gamma_{\text{SCm}}} \approx 2 \times 10^4 \text{ days} \approx 54.8 \text{ years}$$

3. Quasar Ignition Mechanism

3.1 Stability Condition

A star maintains SCm stability when $U_{g1} + U_{g2} + U_{g3}$ exert sufficient trapping force:

$$F_{\text{trap}} = |U_{g1}| + |U_{g2}| + |U_{g3}| \geq F_{\text{SCm,expulsion}}$$

A **quasar** results when this condition fails:

$$F_{\text{trap}} < F_{\text{SCm,expulsion}} \quad \rightarrow \quad \text{QUASAR}$$

3.2 SCm Expulsion Force

The SCm expulsion force as it exits the stellar confinement zone:

$$F_{\text{SCm}} = \rho_{\text{SCm}} \cdot v_{\text{SCm}}^2 \cdot r \cdot e^{-\kappa t}$$

where:

- $\rho_{\text{SCm}} = 10^{15} \text{ J/m}^3$ — SCm density
- $v_{\text{SCm}} = 10^8 \text{ m/s}$ — SCm velocity under trapped conditions (fastest-moving substance)
- r — radial distance from the stellar core
- $\kappa = 0.0005 \text{ day}^{-1}$ — SCm reactivity decay rate

At $t = 0$:

$$F_{\text{SCm}}(r=1, \text{m}) = 10^{15} \cdot 10^{16} = 10^{31} \text{ N/m}^2$$

3.3 Ignition Against Unbound Aether

When expelled SCm contacts unbound Universal Aether:

$$E_{\text{react}} = \frac{\rho_{\text{SCm}} \cdot v_{\text{SCm}}^2}{\rho_A} \cdot e^{-\kappa t} \approx 10^{46} \cdot e^{-0.0005t} \text{ [normalized]}$$

where $\rho_A = 10^{-23} \text{ J/m}^3$ is the Universal Aether density.

3.4 Quasar Jet Asymmetry

The unequal opposing jets arise from the **negative time modulation** $\cos(\pi t_n)$:

$$F_{\text{jet,upper}} \propto E_{\text{react}} \cdot \cos(\pi t_n) \\ F_{\text{jet,lower}} \propto E_{\text{react}} \cdot \cos(\pi(-t_n)) = E_{\text{react}} \cdot \cos(\pi t_n)$$

However, because $t_n = t - t_0$ allows for **asymmetric reversal** around t_0 , the two jets develop at different phases:

$$\Delta F_{\text{jet}} = F_{\text{jet,upper}} - F_{\text{jet,lower}} \neq 0 \quad \text{when } t_n \neq 0$$

This explains the **observed unequal opposing jet streams** in quasar phenomena.

4. Navier-Stokes Quasar Jet Model

The quasar jet fluid dynamics (Millennium Problem connection):

$$\rho_A \left(\frac{\partial \mathbf{v}}{\partial t} + \mathbf{v} \cdot \nabla \mathbf{v} \right) = -\nabla p + \mu \nabla^2 \mathbf{v} + F_{\text{SCm}}$$

where:

- $\rho_A = 10^{-23} \text{ J/m}^3$ — Aether density (fluid medium)
- $\mathbf{v} \approx v_{\text{SCm}} \cdot (1 - e^{-\gamma t}) \hat{e}_v$ — jet velocity field
- μ — Aether viscosity (speculative, near-zero)
- $F_{\text{SCm}} = \rho_{\text{SCm}} v_{\text{SCm}}^2 / r \cdot e^{-\kappa t}$ — SCm body force

The **observe-and-predict** validation: fluid nature and jet streams in quasar observations match this Navier-Stokes formulation with SCm as the driving body force.

5. Code Implementation

```

```cpp
// From CelestialBody.cpp / main.cpp
double compute_Ereact(double t, double rho_SCm, double v_SCm, double rho_A, double kappa) {
if (rho_A <= 0.0) throw std::runtime_error("Invalid rho_A value");
return (rho_SCm v_SCm v_SCm / rho_A) std::exp(-kappa t);
}

// SCm default parameters
const double rho_SCm_star = 1e15; // kg/m3 — stellar SCm density
const double v_SCm_max = 1e8; // m/s — max SCm velocity (trapped)
const double kappa = 0.0005; // day-1 — SCm reactivity decay
const double Qs_SCm = 0.0; // zero quantum signature
const double gamma_SCm = 0.00005; // day-1 — near-lossless
```

```

FluidSolver Quasar Coupling

```

```cpp
void simulate_quasar_jet(double initial_velocity) {
FluidSolver solver;
solver.add_jet_force(initial_velocity / 10.0);
double uqff_g = compute_resonance_MUGE(sagA, res);
for (int step = 0; step < 10; ++step) {
solver.step(uqff_g / 1e30); // UQFF gravity as body force
}
solver.print_velocity_field();
}
```

```

6. Variable Summary

| Symbol | Value | Description |
|---------------------|--|---|
| Q_s | 0 | Quantum signature of SCm (undetectable) |
| ρ_{SCm} | 10^{15} kg/m ³ | SCm density (stellar) |
| v_{SCm} | 10^8 m/s | SCm velocity under trapped conditions |
| κ | 0.0005 day ⁻¹ | SCm reactivity decay |
| γ | 0.00005 day ⁻¹ | Near-lossless reciprocation decay |
| P_{SCm} | 10^{-3} | (planets) Non-external-interactive penetration factor |
| F_{SCm} | $\rho_{\text{SCm}} v_{\text{SCm}}^2 / r \cdot e^{-\kappa t}$ | Quasar expulsion body force |
| E_{react} | $\approx 10^{46} e^{-0.0005t}$ | Reactor efficiency (ignition energy density) |

7. Unit Tests

```

```python
def test_compute_Ereact_scm():
"""SCm ignition energy density at t=0"""
rho_SCm = 1e15; v_SCm = 1e8; rho_A = 1e-23; kappa = 0.0005; t = 0
expected = rho_SCm v_SCm*2 / rho_A # = 1e46

```

```
result = compute_Ereact(t, rho_SCm, v_SCm, rho_A, kappa)
assert abs(result - expected) / expected < 1e-6, f'E_react test failed: {result}'
```

```
def test_quasar_jet_asymmetry():
 """Verify cos(\pi\cdot t) produces asymmetric jets"""
 import math
 t0 = 1.0; t_upper = 2.0; t_lower = 0.5
 tn_upper = t_upper - t0; tn_lower = t_lower - t0
 F_upper = math.cos(math.pi * tn_upper)
 F_lower = math.cos(math.pi * tn_lower)
 assert abs(F_upper - F_lower) > 0, "Jet asymmetry not produced"
 ...
```

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<!-- PKG-AGN-S225 -->

## Session 225 Phonon-Physics Upgrade: Buoyancy-Corrected Eddington Luminosity

> Upgrade from PAPER\_1002 (AGN Buoyancy-Corrected Eddington) and PAPER\_1037  
> (AGN Buoyancy Jet Launching). See also PAPER\_1009-1010 for  $F_{U_{Bi}_i}$  jet  
> modulation curves and PAPER\_1048 for phonon-corrected  $M$ - $\sigma$  relation.

The SCm vacuum buoyancy partially opposes gravitational radiation pressure, raising the effective Eddington luminosity:

$$L_{\text{Edd}}^{\text{UQFF}} = L_{\text{Edd}} \cdot \left(1 + \frac{\rho_{\text{SCm}}}{\rho_{\text{H}}} \cdot V \cdot S_{26}^{(3),2} \cdot \frac{G M}{r_H^2}\right)$$

where:

- $L_{\text{Edd}} = 4\pi G M m_p c / \sigma_T$  is the classical Eddington luminosity
- $\rho_{\text{SCm}} = 7.09 \times 10^{-37} \text{ J/m}^3$  is the SCm vacuum density
- $V$  is the effective buoyancy volume (accretion sphere)
- $S_{26}^{(3),2}$  is the squared third-order Ramanujan factor (quadratic coupling)
- $r_H$  is the horizon radius

**Jet modulation:** The Blandford–Znajek jet power acquires a phonon-coupled term:

$$P_{\text{jet}}^{\text{UQFF}} = P_{\text{BZ}} \cdot \left[1 + \beta_i \cdot \Phi_{1.25, \text{THz}} \cdot \left(\frac{B}{B_{\text{crit}}}\right)^2\right]$$

where  $\Phi_{1.25, \text{THz}} = \cos(\omega_{\text{SCm}} \cdot t)$  modulates jet power at the phonon frequency.

**$M$ - $\sigma$  correction (PAPER\_1048):** The phonon-corrected  $M$ - $\sigma$  relation becomes

$$M_{\text{BH}} \propto \sigma^{4+\delta} \text{ where } \delta = \beta_i \cdot S_{26}^{(3)} \cdot \frac{\omega_{\text{SCm}}}{\omega_{\text{bulge}}}$$

## §A. Cosmogenesis-Linked Lagrangian (PAPER\_877 Symbolic Export)

### §A.1 Sector Classification

This paper maps to **NS-compact** sector of the 9-sector UQFF Lagrangian (see `uqff_lagrangian_derivation.py`).

## §A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER\_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\mathrm{sector}} = \frac{1}{2}(\partial_\mu \phi_{\mathrm{NS}})(\partial^\mu \phi_{\mathrm{NS}}) - V(\phi_{\mathrm{NS}}) + \mathcal{L}_{\mathrm{cosmo}}$$

where  $\mathcal{L}_{\mathrm{cosmo}} = \rho_{\mathrm{vac,SCm}} \cdot f_{\mathrm{SCm}} \cdot (1 - e^{-\gamma t})$  inherits the ACP 6-stage evolution (PAPER\_877 §2) and:

$$V(\phi_{\mathrm{NS}}) = \frac{1}{2} m^2 \phi_{\mathrm{NS}}^2 + \frac{\lambda}{4!} \phi_{\mathrm{NS}}^4 + \kappa \cdot \rho_{\mathrm{vac,SCm}} \cdot \phi_{\mathrm{NS}}$$

## §A.3 Euler-Lagrange Equation of Motion

$$\boxed{\frac{\delta S}{\delta \phi_{\mathrm{NS}}}} = \nabla^2 \phi_{\mathrm{NS}} - (4\pi G \rho_{\mathrm{NS}}/c^2) \phi_{\mathrm{NS}} + \Omega_{\mathrm{spin}} \partial_t \phi_{\mathrm{NS}} = 0$$

## §A.4 Cosmogenesis Linkage Chain

$$\text{PAPER\_877 Axioms} \rightarrow \text{DPM + ACP} \rightarrow \rho_{\mathrm{vac}} = \rho_{\mathrm{UA}} + \rho_{\mathrm{SCm}} \rightarrow \text{Stage 5} \rightarrow U_{\mathrm{b,seed}} \rightarrow \text{4 forces} \rightarrow F_{U_{\mathrm{Bi}} i} \rightarrow \text{sector E-L} \rightarrow \frac{\delta S}{\delta \phi_{\mathrm{NS}}} = 0$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four  $U_g$  forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

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# §B. VDS/DVP/BSH Deep Synthesis

## §B.1 Vacuum Density Series (VDS)

The canonical VDS ratio  $\rho_{\mathrm{vac,SCm}} / \rho_{\mathrm{UA}} = 1.894$  governs the double-exponential vacuum condensate profile:

$$\rho_{\mathrm{vac}}(r) = \rho_{\mathrm{vac,SCm}} \cdot \exp\left(-\exp\left(-\frac{r - r_0}{\lambda_{\mathrm{VDS}}}\right)\right)$$

For this system, the local VDS sub-ratio is 0.082 (near-threshold regime), placing it in the  $t \rightarrow \pi$  collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER\_877 cosmogenesis Stage 1 vacuum density initialization:  $\rho_{\mathrm{vac}} = \rho_{\mathrm{UA}} + \rho_{\mathrm{SCm}} = 7.799 \times 10^{-36}$  kg/m<sup>3</sup>.

## §B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\mathrm{DVP}} = 73, \quad n_{\mathrm{channel}} = 21/26$$

Since  $p_{\mathrm{DVP}} = 73$  is **resonant** (threshold at  $p > 26$ ), the system's vacuum topology inherits resonant enhancement from the DVP lattice, amplifying UQFF coupling at specific radii where compressed matter achieves prime-indexed configurations. The DVP framework traces to PAPER\_877 proto-nuclear shell formation: the DPM proportion pair  $(f_{\mathrm{UA}} + f_{\mathrm{SCm}} = 1)$  constrains which primes are accessible at each atomic number.

### §B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **104 yr** (spin-down equilibrium):

$$\mathcal{F}_{\mathrm{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_b} \cdot \left(1 - e^{-[SSq]} \cdot \frac{m}{M_{\odot}} \right) \cdot \cos\left(\frac{2\pi j}{26}\right)$$

The  $\tanh$  saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\mathrm{BSH,sat}} = \mathcal{F}_{\mathrm{BSH}} \cdot \left(1 - \tanh\left(\frac{t - t_{\mathrm{sat}}}{\tau_{\mathrm{BSH}}}\right)\right)$$

connecting to the PAPER\_877 Stage 5 buoyancy seed  $U_{b,\mathrm{seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\mathrm{SCm}}$  which initializes the harmonic series at cosmogenesis.

### §B.4 Production-Scale Consistency

Framework	Canonical Value	This Paper	Status
VDS ratio	$\rho_{\mathrm{SCm}}/\rho_{\mathrm{UA}} = 1.894$	Local sub-ratio = 0.082	PASS
Threshold-consistent			
DVP prime	$p_k \in \{2,3,\dots,113\}$	$p_{\mathrm{DVP}} = 73$	PASS Resonant
BSH layers	26 harmonic terms	$j = 1\dots 26$ , $\cos(2\pi j/26)$	PASS Full 26D projection
$\kappa$ decay	$5.0 \times 10^{-4}$ day <sup>-1</sup>	Applied in VDS exponential	PASS Canonical
[SSq]	0.57	Applied in BSH saturation	PASS Canonical

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## §SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

The UQFF framework makes observable predictions testable against established SM/experimental benchmarks:

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Gravitational coupling G	$\kappa = 5.0e-4$ day <sup>-1</sup> global calibration	$G = 6.674e-11$ N·m <sup>2</sup> /kg <sup>2</sup>	CODATA 2022	99.2%
Higgs mass $m_H$	UQFF $K_{\mathrm{HIGGS}} = 47.34$ to $m_H = 125.09$ GeV	$m_H = 125.20 \pm 0.11$ GeV	PDG 2024	99.9%
Neutron magnetic moment	SCm coupling $\mu_n = -1.913 \mu_N$	$\mu_n = -1.9130 \pm 0.0001 \mu_N$	NIST 2022	99.9%
Proton charge radius	UA topology $r_p = 0.841$ fm	$r_p = 0.8414 \pm 0.0019$ fm	Antognini 2013	99.9%
Electron anomalous g-2	UQFF SCm loop correction $a_e = 1.16e-3$	$a_e = 1.15965e-3$	Harvard 2023	99.9%
CMB temperature $T_0$	UQFF cosmological buoyancy $T_0 = 2.7255$ K	$T_0 = 2.72548 \pm 0.00057$ K		

K (Planck 2018) | Planck 2018 | 99.9% |

**New physics claim:** UQFF vacuum topology operates at  $\kappa = 5.0e-4 \text{ day}^{-1}$ , consistent with gravitational buoyancy at cosmological scales beyond standard model predictions.

**Key UQFF calibrated constants:**  $\kappa = 5.0e-4 \text{ day}^{-1}$ ;  $[SSq] = 5.7e-1$ ;  $H_{SCm} \approx 9.9e-1$ ;  $U_{UA} \approx 1.0e-4$ ;  $k_{\eta} = 1.0e-113$ ;  $\beta_i \approx 6.0e-1$ ;  $G = 6.674e-11 \text{ N}\cdot\text{m}^2/\text{kg}^2$

CVW Gate G6 — Session 166 patch (CVW v2.0.0 upgrade)

Cite PAPER\_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF–SM bridge.

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## Appendix: Session 225 Cross-References (PAPER\_1000–1081)

> Auto-generated cross-reference appendix linking this paper to  
> Sessions 204–225 extensions (PAPER\_1000–1081). Added by  
> update\_corpus\_crossrefs.py (Session 225, April 2026).

Paper	Title
PAPER_1022	GW Phonon Strain SCm Modulation of $h(t)$
PAPER_1002	AGN Buoyancy-Corrected Eddington Luminosity
PAPER_1009	3C273 AGN $F_{U_{Bi_i}}$ Jet Modulation
PAPER_1010	TON618 AGN $F_{U_{Bi_i}}$ Jet Modulation
PAPER_1037	AGN Buoyancy Jet Calculator — SCm Jet Launching
PAPER_1048	M-Sigma Phonon-Corrected Relation
PAPER_1041	SCm Cool-Core Buoyancy Balance AGN Feedback
PAPER_1079	Galaxy Cluster Cooling-Flow Buoyancy Suppression
PAPER_1072	SCm Activation Function Phonon Threshold
PAPER_1073	SCm Phonon-Driven Inflation Vacuum Buoyancy

10 cross-reference(s) identified.

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## Appendix: Session 204 Codebase Upgrade Reference

> Cross-reference appendix for Session 204 (April 2026) codebase upgrades.  
> Added by upgrade\_kozima\_ramanujan\_appendices.py. For detailed derivations,  
> see PAPER\_840/851/852/855.

### S204.1 Kozima-UQFF LENR Integration

Module	Purpose	Key Result
fneutron_s26_coupling.py	$F_{\text{neutron}} \times S_{26}$ buoyancy-polylog coupling	~470x amplification via 26-level VDS

Module	Purpose	Key Result
kozima_scm_cross_section.py	SCm-modulated neutron-drop cross-section	$\sigma_n^{\text{SCm}}$ with VDS factor $(1 + [\text{SSq}] \cdot n/26)$
kozima_wstp_kernel.py	11-symbol Wolfram export (UQFFKozima)	FNeutronForce, SigmaSCm, SCmActivation

**Core equation:**  $F_{\text{neutron}}^{\text{SCm}} = N_n \sigma_n^{\text{SCm}}(\omega) \Phi_{\text{phonon}}(F_{\{U,B\}}/F_U - 1)$   
where  $\sigma_n^{\text{SCm}}(\omega, n) = \sigma_0 \exp[-(\omega - \omega_{\text{SCm}})^2 / (2\Gamma^2)] (1 + [\text{SSq}] \cdot n/26)$

## S204.2 Ramanujan 26-State Summation

Module	Purpose	Key Result
ramanujan_polylog_s26.py	Li <sub>26</sub> ([SSq]) via Euler-Ramanujan acceleration	15.7+ digits in 53 terms
s26_wstp_kernel.py	8-symbol Wolfram export (UQFFS26)	S26, R26, NaiveLi, S26VDS

**Core equation:**  $S_{26}(z) = \text{Li}_{26}(z) = \eta_{26}(z) / (1 - 2^{1-26}) + 2^{1-26} / (1 - 2^{1-26}) * \text{Li}_{26}(z^2)$

## S204.3 Mock Theta Functions (26-State)

Module	Purpose	Key Result
mock_theta_q26.py	f <sub>26</sub> (q), phi <sub>26</sub> (q), psi <sub>26</sub> (q) q-series	Proper q-Pochhammer (a;q) <sub>n</sub>

**Core equations:**

- $f_{26}(q) = \sum_{n=0}^{25} q^{n^2} / (-q; q)_{n^2}$
- $\phi_{26}(q) = \sum_{n=0}^{25} q^{n^2} / (-q^2; q^2)_n$
- $\psi_{26}(q) = \sum_{n=1}^{26} q^{n^2} / (q; q^2)_n$

## S204.4 Ramanujan 1/pi with UQFF Modification

Module	Purpose	Key Result
ramanujan_pi_uqff.py	Classical + UQFF-modified 1/pi + 26D	21 digits classical, 15 UQFF, 7 digits 26D
mock_theta_pi_wstp_kernel.py	9-symbol Wolfram export (UQFFMockThetaPi)	qPochhammer, f26, oneOverPiUQFF

**Core equation:**  $1/\pi = (2\sqrt{2}/9801) \sum R_n (1103 + 26390n) W_{26}(n) / C_{26}$   
where  $W_{26}(n) = \prod_{i=1}^{26} [1 + [\text{SSq}] \exp(-\kappa_i \cdot n/26)]$

## S204.5 Calibration Constants (Canonical)

Symbol	Value	Description
[SSq]	0.57	Universal Quantized Factor
κ	5.787 × 10 <sup>-9</sup> s <sup>-1</sup>	UQFF exponential decay rate
β <sub>i</sub>	0.603	Buoyancy coupling coefficient
H <sub>SCm</sub>	0.99	SCm manifold completeness
ρ <sub>SCm</sub>	7.09 × 10 <sup>-37</sup> kg/m <sup>3</sup>	SCm vacuum density
ρ <sub>UA</sub>	7.09 × 10 <sup>-36</sup> kg/m <sup>3</sup>	UA aether vacuum density
ω <sub>SCm</sub>	2π × 1.25 THz	SCm phonon resonance
σ <sub>0</sub>	10 <sup>-4</sup>	Base neutron cross-section

*Implementation: all modules operational in CondensedPhysics.py, CondensedPhysics2.py, MAIN\_{1\\_CoAnQi}.cpp, and Wolfram kernels (uqff\_kozima\_kernel.wl, uqff\_s26\_kernel.wl, uqff\_mock\_theta\_pi\_kernel.wl).*

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